

Design Automation) tools, knowledge of memory chip architecture, analogue and digital design techniques, etc. are required. Software skills such as programming in C, C++, Java, etc. are most hunted for.

Besides this, soft skills—communication skills, team spirit, and so on are also assessed. The industry makes efforts to reach potential talents by accessing the campuses of universities through panel discussions, seminars, presentations, internships, campus interviews, and sponsored programs.

The Necessity Of Training

Training is a crucial aspect in this field. After recruitment, employees undergo training that might be basic induction training, all the way to high-end training in embedded systems tools and technologies.

“Freshly-recruited candidates undergo training and workshops on embedded technologies and communication protocols. The training period ranges anywhere between three to six months depending on the domain specialisation.

“These candidates are deployed into projects immediately after completion of training,” informs Anup Sable, vice president—Automotive and Embedded Line of Business, KPIT Cummins Infosystems Ltd, which provides solutions and services for hardware design and software for embedded systems.

Why we mention training is because companies want embedded systems professionals to work with the best available technologies and EDA tools to innovate products. With the trend of finishing and training schools cropping up, a candidate can get registered at the schools while pursuing their regular degrees. At these schools—private and government—short-term courses are provided to enable students to get hands-on experience with the tools and applications used in the industry.

“Training of whatever duration is important since it is different from conventional, commercial application development. Polishing of skills and upgrading of technology knowledge is required due to the emergence of new standards, platforms, and development tools” avers Sanjeet Banerji, senior vice president—Telecom and Embedded Systems, Datamatics Ltd, which provides a wide range of solutions and services, including those for the embedded domain.

The Growth Path And Remuneration

The exact designations differ from company to company. An aspirant can choose either to be on the hardware front or on software development. It is very important to maintain a balance between the two, as the embedded control software works closely with the microprocessor and microcontroller.

There are separate teams for hardware and software groups. The stages of the growth path can be divided into entry-level, senior-level, and top-level.



A consortium of companies is actively making efforts to bridge the widening gap between requirement and availability of highly skilled and trained engineers”

P V G Menon
Senior business development manager
CTO Library & IP Partnership
NXP Semiconductors

■ Entry-Level

At this level, fresh recruits are assigned the tasks of testing, validation, design, and such. They are typically designated “trainee engineers” or “trainee testers.”

During this stage, one needs to apply the knowledge, basics, and fundamentals of embedded systems to get hands-on skills of working with industry-standard tools and technologies.

The employee is physically involved in combining hardware, software and other mechanical parts using EDA tools to make a product for a specified purpose. Intensive learning takes place at this stage; senior-level engineers act as guides during the process of developing a product.

The time taken to advance to the next level ranges from two to four years.

Remuneration: Rs 2 to 2.5 lakh Per Annum.

■ Senior-Level

Here, one is referred to as “engineer,” “senior engineer,” “team leader,” or “principal architect.” They are in charge of the entry-level engineers and the developments taking place under current projects. A senior design engineer leads a group of entry-level engineers working on specific areas of development in a particular project.

The time taken to advance to the next level depends entirely upon the contribution to the projects and the results derived, but an indicative figure would be three to four years.

Remuneration: Rs 3 to 4 lakh P.A.

■ Top-Level

When one reaches here, the individual can choose a track either on the technical side—such as “project lead”—or the management side, as a “project manager.” The top-level consists of project leaders, project managers, and general managers. The project leader is the incharge of a project, and keeps track of project developments. Project managers are involved with client relations and quality management. They check the progress of the projects and report to the general manager, who takes the strategic and managerial decisions related to the projects.

Remuneration: Rs 6 lakh and above P.A.

Wrapping It Up

According to a report—*Indian Semiconductor Industry and its Impact*, by The ISA (Indian Semiconductor Association) and Frost & Sullivan, there would be a requirement for 7,81,780 engineers in the semiconductor and embedded design verticals. Universities do provide formal education, but changing curricula in tune with rapidly-changing technologies and trends of the industry verticals is hard.

This field is for innovators: those who can take up challenges and develop new products and applications. A league of smart brains is behind building your cell phone, washing machine, your television. You can choose to be a passive user of those smart devices—or be the innovator of smarter ones! ■

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